

Conservation-constraint load d_c predicts physics-reasoning failures

a countable physics-structure variable links item difficulty, formulation support, and exact-graded validation

1 Count conservation constraints



elastic collision

momentum + energy

$$d_c = 2$$

2 Per-constraint penalty

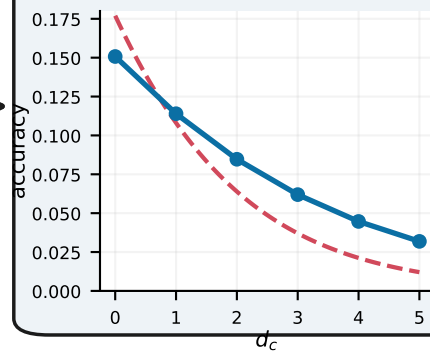
$$p \times p \times p$$

correct only if all close

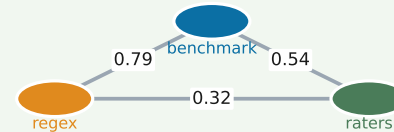
$$\text{logit } P \propto d_c$$

OR ≈ 0.68 per added constraint

3 Failures rise with d_c



4a Labels are reproducible



4b Formulation support carries the rescue

laws: +0.016 equations: monotone

exact-graded probe: no judge

reproducible item-level variable for benchmark design and equation scaffolding